

ENV 790.30 - Time Series Analysis for Energy Data | Spring 2025

Assignment 3 - Due date 02/03/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A03_Sp25.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

Please keep this R code chunk options for the report. It is easier for us to grade when we can see code and output together. And the tidy.opts will make sure that line breaks on your code chunks are automatically added for better visualization.

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

Questions

Consider the same data you used for A2 from the spreadsheet "Table_10.1_Renewable_Energy_Production_and_Consumption". The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. This time you will work only with the following columns: **Total Renewable Energy Production**; and **Hydroelectric Power Consumption**.

```
#Importing data set
library(readxl)
library(openxlsx)
getwd()
```

```
## [1] "/home/guest/ZZhang727"
```

```
#Importing data set without change the original file using read.xlsx
energy_data1 <-
  read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx",
             skip = 12, sheet="Monthly Data", col_names=FALSE)
```

```
## New names:
## * ' ' -> '...1'
## * ' ' -> '...2'
```

```
## * '' -> '...3'
## * '' -> '...4'
## * '' -> '...5'
## * '' -> '...6'
## * '' -> '...7'
## * '' -> '...8'
## * '' -> '...9'
## * '' -> '...10'
## * '' -> '...11'
## * '' -> '...12'
## * '' -> '...13'
## * '' -> '...14'
```

#Now let's extract the column names from row 11

```
read_col_names <-
  read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx",
             skip = 10,n_max = 1, sheet="Monthly Data",col_names=FALSE)
```

New names:

```
## * '' -> '...1'
## * '' -> '...2'
## * '' -> '...3'
## * '' -> '...4'
## * '' -> '...5'
## * '' -> '...6'
## * '' -> '...7'
## * '' -> '...8'
## * '' -> '...9'
## * '' -> '...10'
## * '' -> '...11'
## * '' -> '...12'
## * '' -> '...13'
## * '' -> '...14'
```

#Assign the column names to the data set

```
colnames(energy_data1) <- read_col_names
```

#Visualize the first rows of the data set

```
head(energy_data1)
```

```
## # A tibble: 6 x 14
##   Month                'Wood Energy Production' 'Biofuels Production'
##   <dtm>                <dbl> <chr>
## 1 1973-01-01 00:00:00      130. Not Available
## 2 1973-02-01 00:00:00      117. Not Available
## 3 1973-03-01 00:00:00      130. Not Available
## 4 1973-04-01 00:00:00      125. Not Available
## 5 1973-05-01 00:00:00      130. Not Available
## 6 1973-06-01 00:00:00      125. Not Available
## # i 11 more variables: 'Total Biomass Energy Production' <dbl>,
## #   'Total Renewable Energy Production' <dbl>,
## #   'Hydroelectric Power Consumption' <dbl>,
## #   'Geothermal Energy Consumption' <dbl>, 'Solar Energy Consumption' <chr>,
```

```
## # 'Wind Energy Consumption' <chr>, 'Wood Energy Consumption' <dbl>,
## # 'Waste Energy Consumption' <dbl>, 'Biofuels Consumption' <chr>,
## # 'Total Biomass Energy Consumption' <dbl>, ...
```

```
#Select two columns: **Total Renewable Energy Production** and **Hydroelectric Power Consumption**
```

```
energy_data_new <- energy_data1[, c(1, 5, 6)]

head(energy_data_new)
```

```
## # A tibble: 6 x 3
##   Month                'Total Renewable Energy Production' Hydroelectric Power ~1
##   <dtm>                                <dbl>                                <dbl>
## 1 1973-01-01 00:00:00                220.                                89.6
## 2 1973-02-01 00:00:00                197.                                79.5
## 3 1973-03-01 00:00:00                219.                                88.3
## 4 1973-04-01 00:00:00                209.                                83.2
## 5 1973-05-01 00:00:00                216.                                85.6
## 6 1973-06-01 00:00:00                208.                                82.1
## # i abbreviated name: 1: 'Hydroelectric Power Consumption'
```

```
nenergy <- ncol(energy_data_new)-1
nobs <- nrow(energy_data_new)

energy_data_tsa <- ts(energy_data_new[,2:3], start=c(1973,1), frequency=12)
```

Create a data frame structure with these two time series only.

R packages needed for this assignment: “forecast”, “tseries”, and “Kendall”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.

```
#Load/install required package here
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

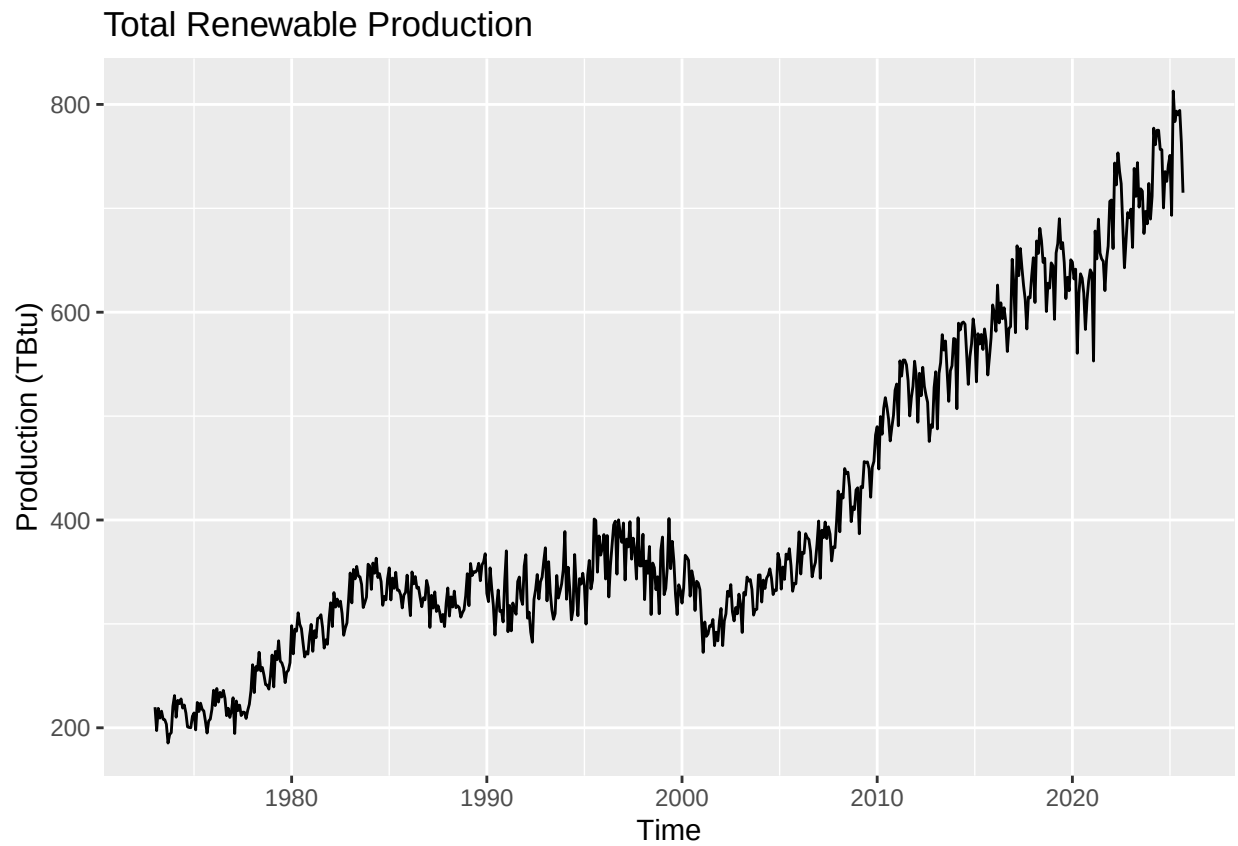
```
library(tseries)
library(Kendall)
library(ggplot2)
```

```
##Trend Component
```

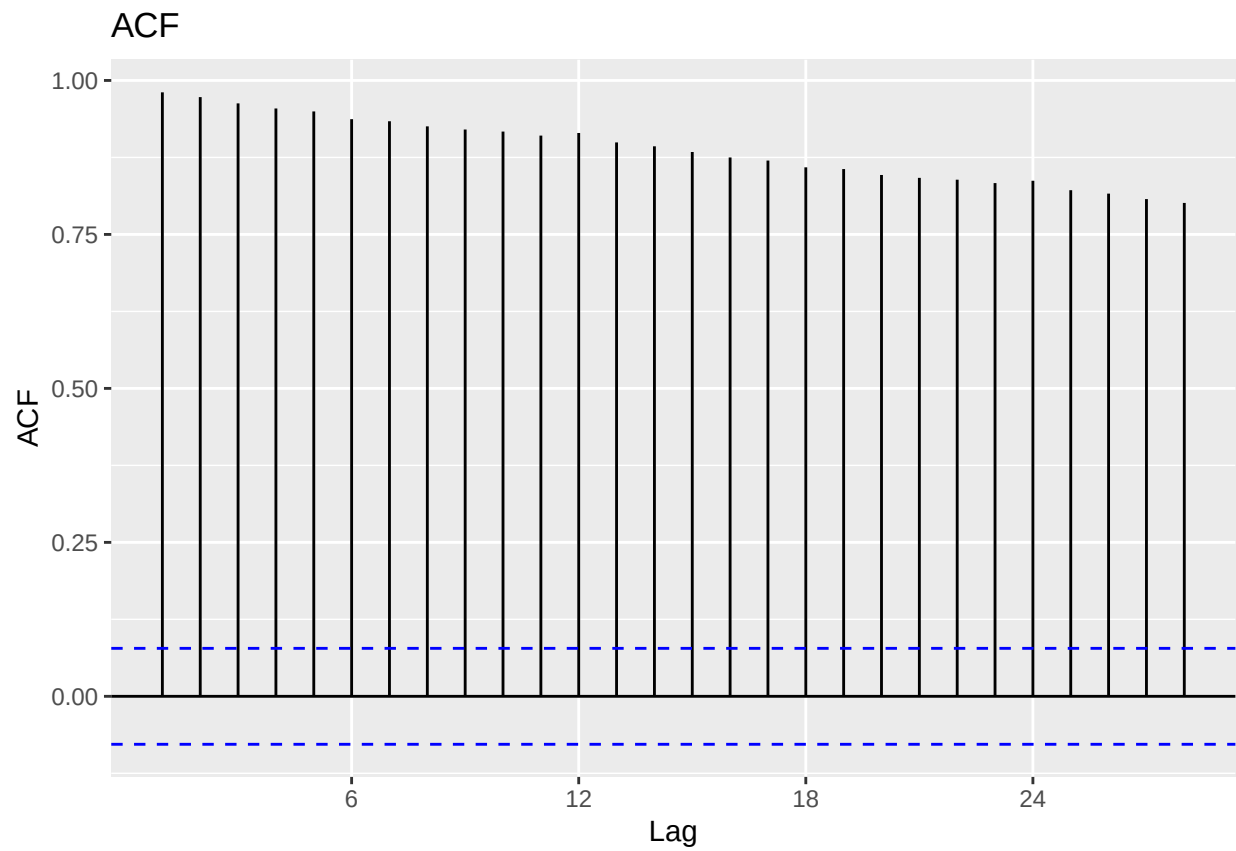
Q1

For each series (Total Renewable Production and Hydroelectric Consumption) create three plots arranged in a row (side-by-side): (1) time series plot, (2) ACF, (3) PACF. Use `cowplot::plot_grid()` to place them in a grid.

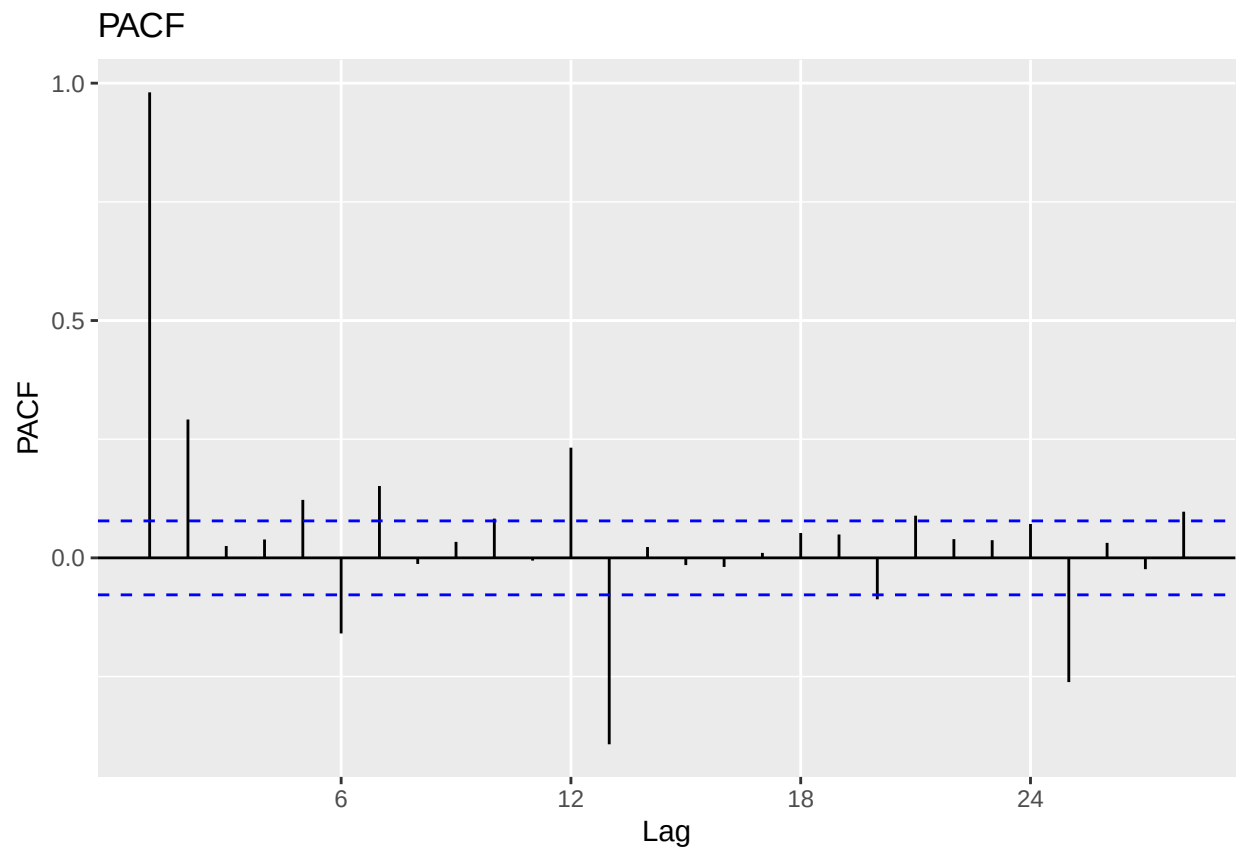
```
#Renewable Production Plots
ts_plot_renewable <- autoplot(energy_data_tsa[,1]) +
  ggtitle("Total Renewable Production")+
  ylab("Production (TBtu)")
ts_plot_renewable
```



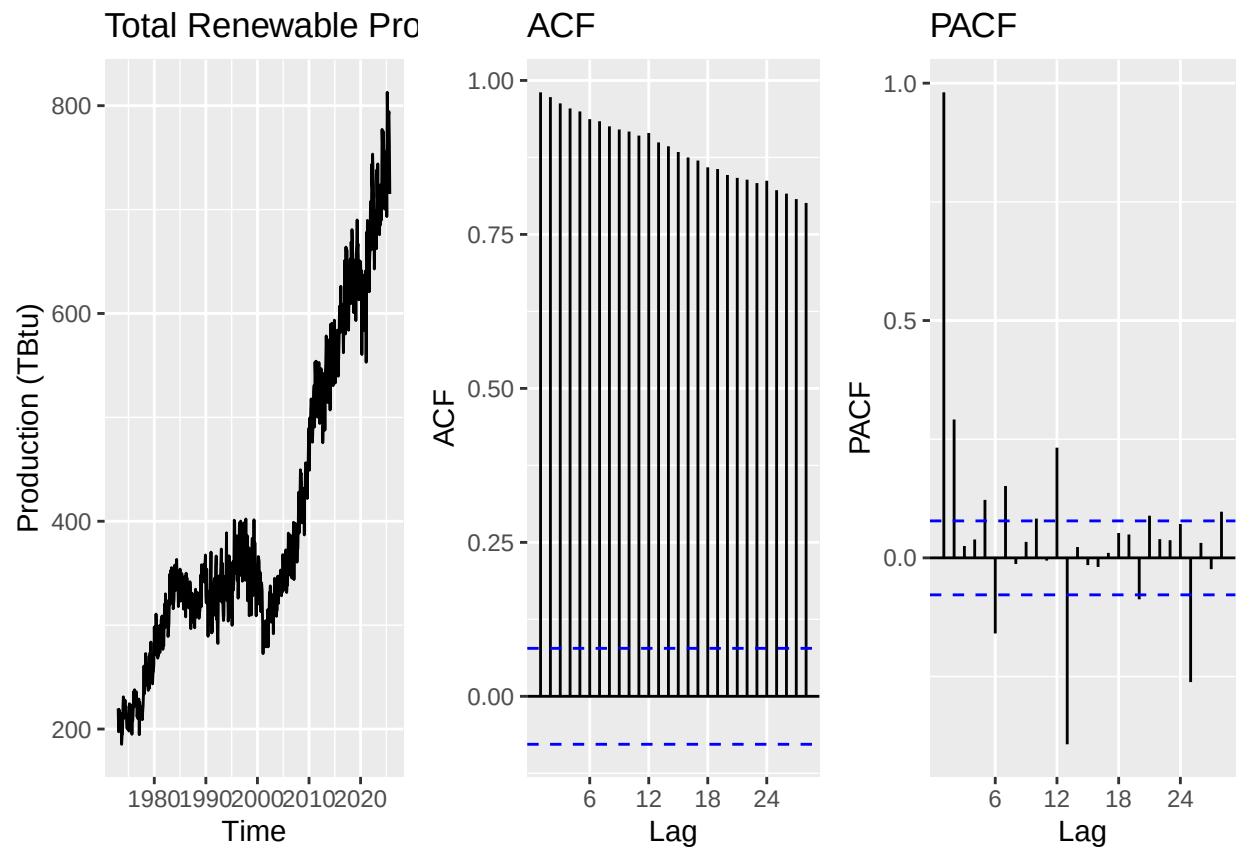
```
Renewable_acf <- ggAcf(na.omit(energy_data_tsa[,1])) + ggtitle("ACF")
Renewable_acf
```



```
Renewable_Pacf <- ggPacf(na.omit(energy_data_tsa[,1])) + ggtitle("PACF")  
Renewable_Pacf
```

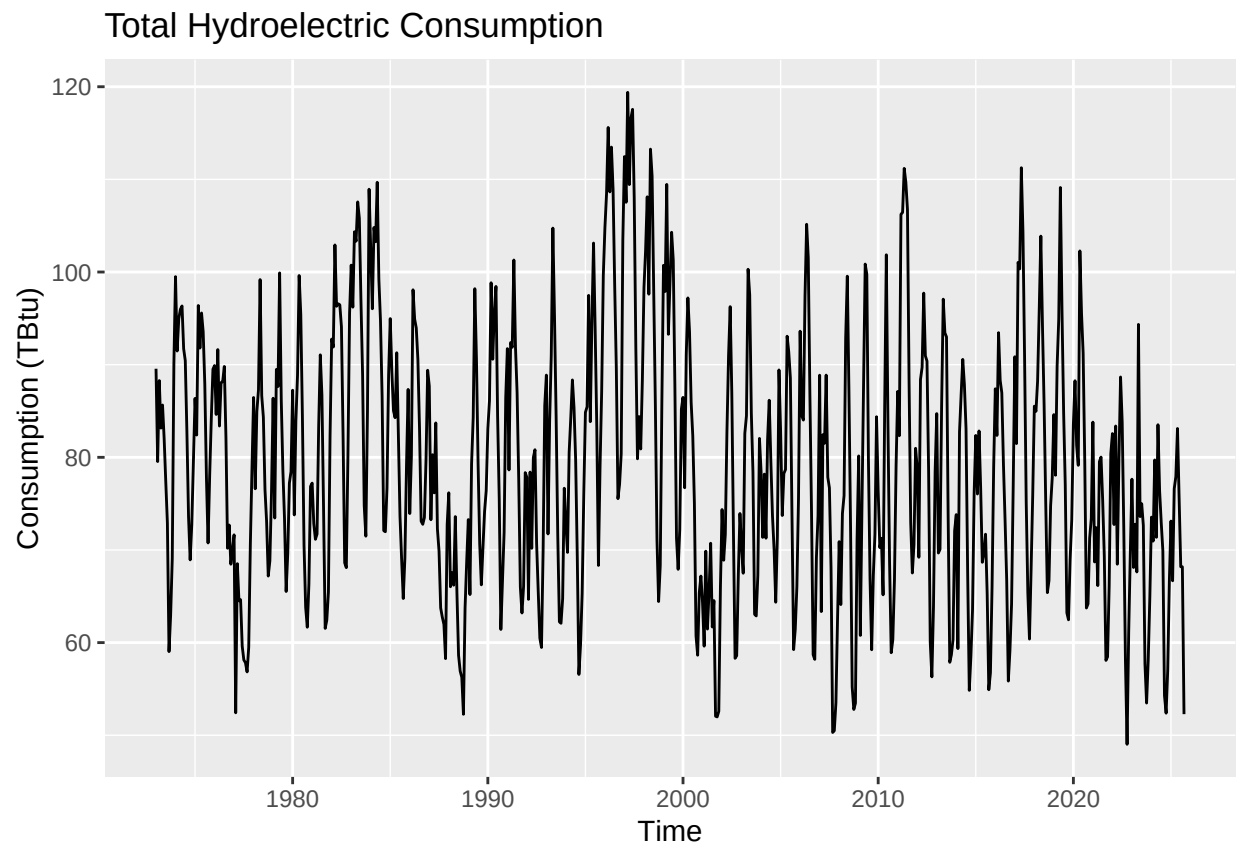


```
Renewable_combined <- cowplot::plot_grid(ts_plot_renewable, Renewable_acf, Renewable_Pacf, nrow = 1)  
Renewable_combined
```

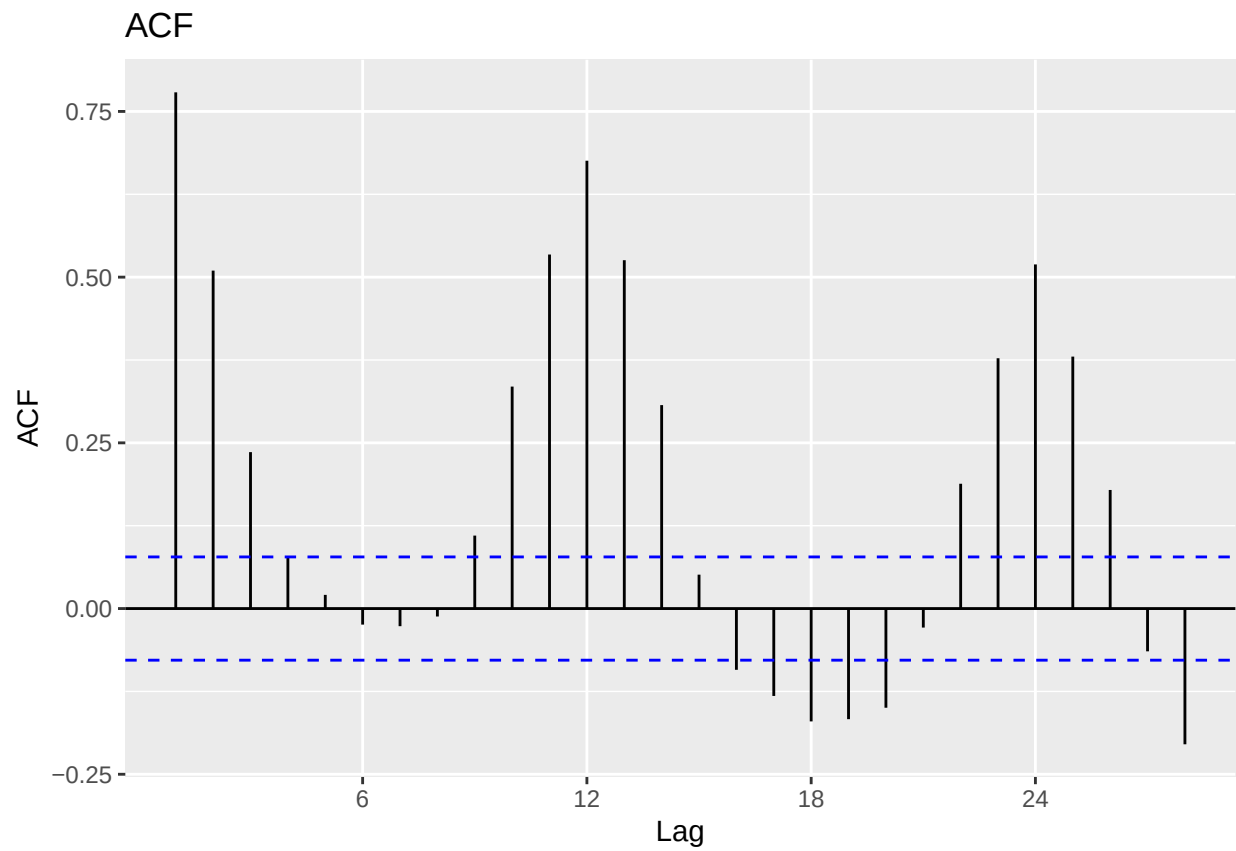


```
library(cowplot)

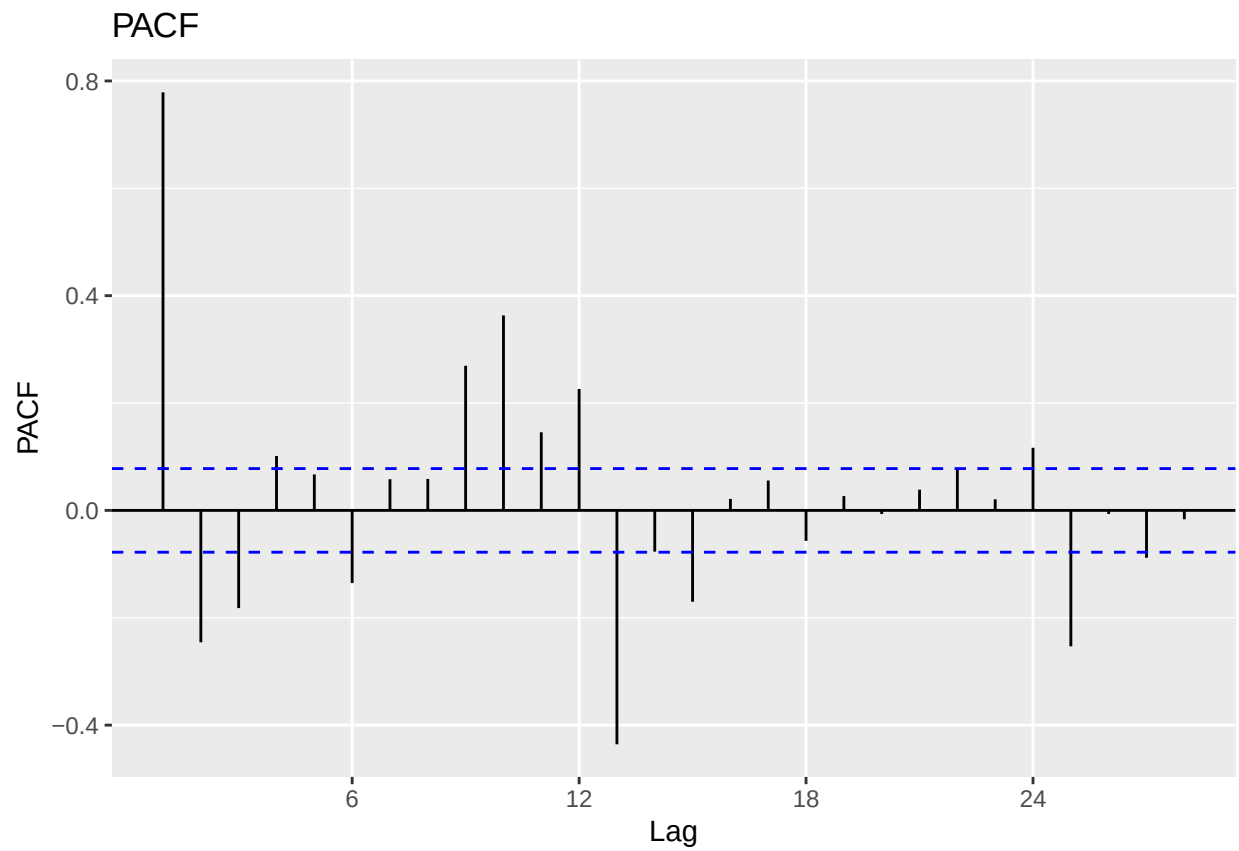
#Hydroelectric Consumption Plots
ts_plot_hydro <- autoplot(energy_data_tsa[,2]) +
  ggtitle("Total Hydroelectric Consumption") +
  ylab("Consumption (TBtu)")
ts_plot_hydro
```



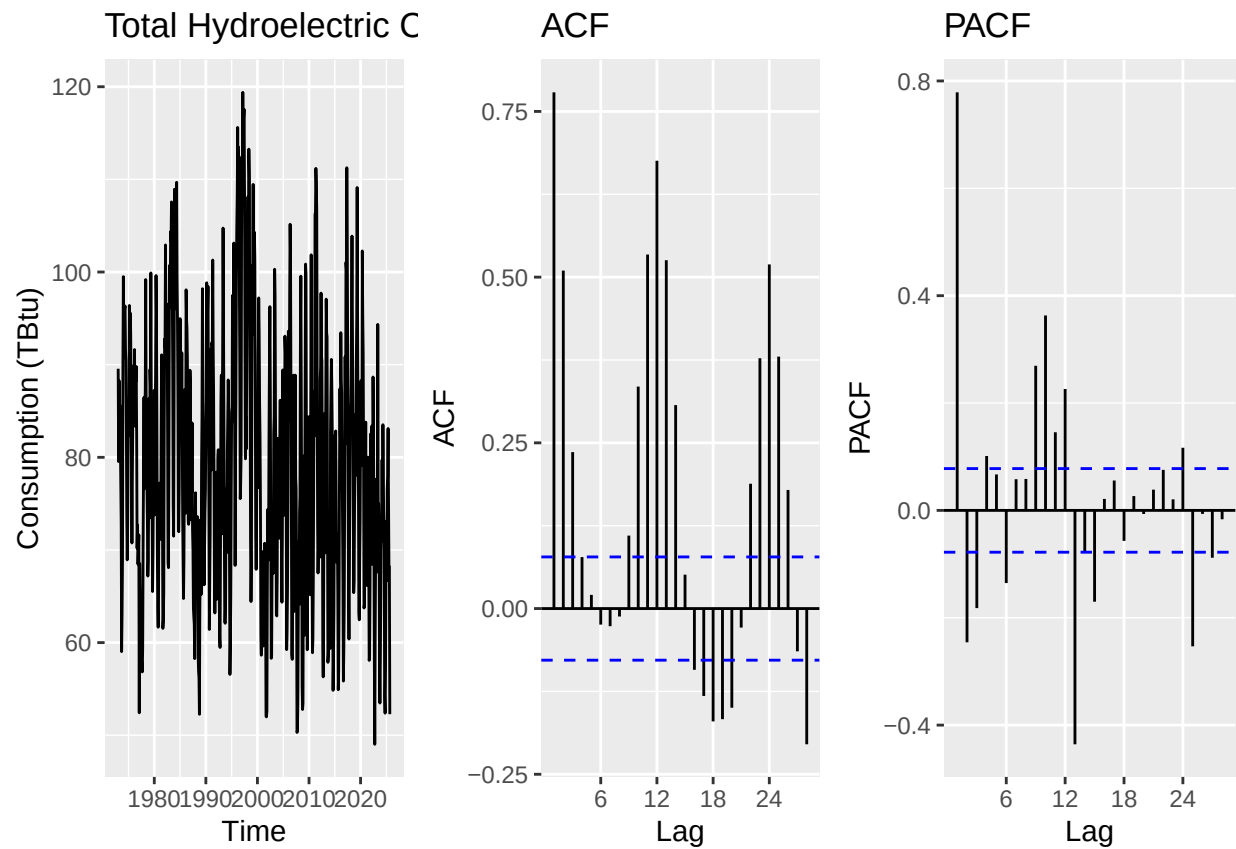
```
Hydro_acf <- ggAcf(na.omit(energy_data_tsa[,2])) +  
  ggtitle("ACF")  
Hydro_acf
```



```
Hydro_Pacf <- ggPacf(na.omit(energy_data_tsa[,2])) +  
  ggtitle("PACF")  
Hydro_Pacf
```

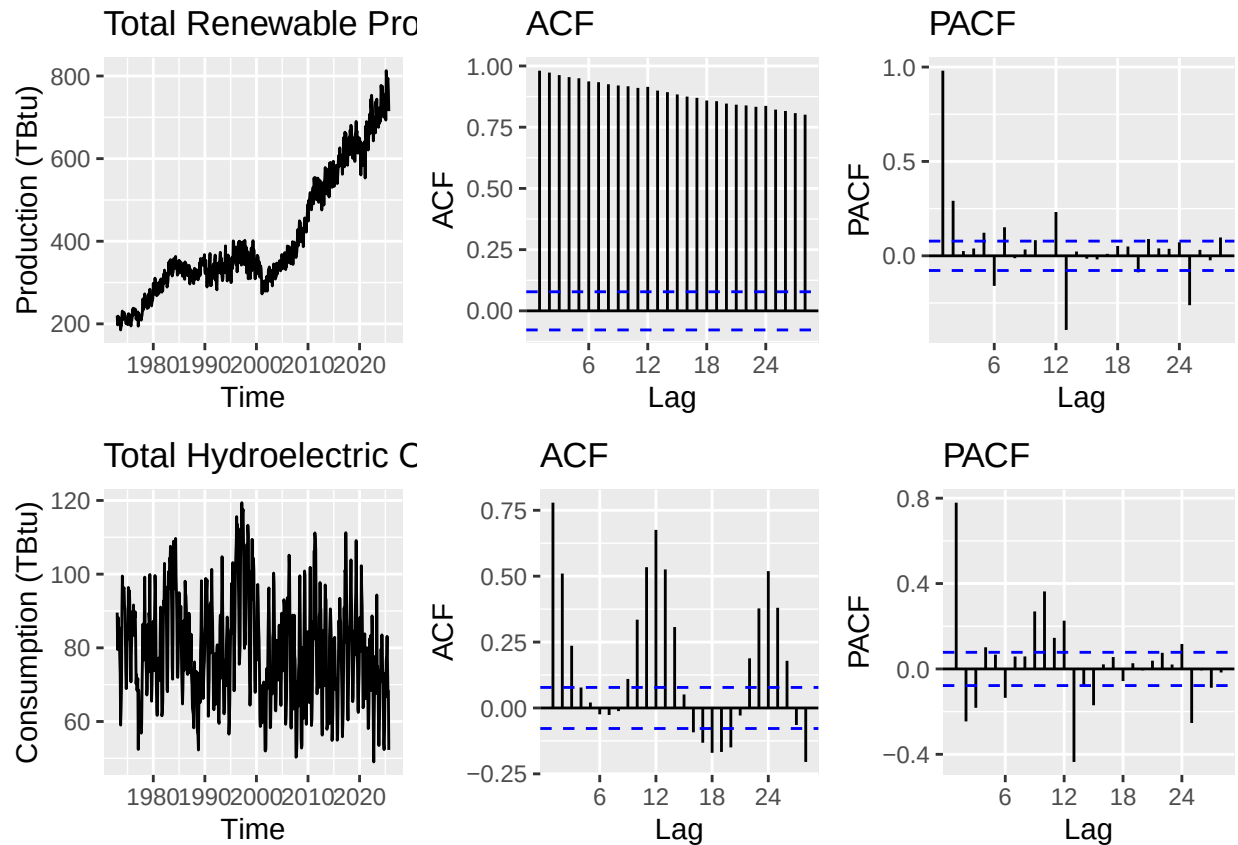


```
Hydro_combined <- plot_grid(ts_plot_hydro, Hydro_acf, Hydro_Pacf, nrow = 1)  
Hydro_combined
```



```
#Combine Renewable and Hydroelectric together
```

```
Combined <-plot_grid(Renewable_combined, Hydro_combined, ncol = 1)
Combined
```



Q2

From the plot in Q1, do the series Total Renewable Energy Production and Hydroelectric Power Consumption appear to have a trend? If yes, what kind of trend?

Answer: The Total Renewable Energy Production shows an trend of increasing in these 40 years, while the Hydroelectric Power Consumption appear to fluctuate around 80 TBtu. The Acf of Total Renewable Energy Production is high & positive,decaying slowly, which could indicate a non-stationary trend. The Acf of Hydroelectric Power Consumption fluctuates between positive and negative around . The Acf of Hydroelectric Power Consumption also increases each 12 lag, showing a seasonality and stationary trend.

Q3

Use the `lm()` function to fit a linear trend to the two time series. Ask R to print the summary of the regression. Interpret the regression output, i.e., slope and intercept. Save the regression coefficients for further analysis.

```
#Create vector t
t <- c(1:nobs)

#Fit a linear trend to TS of renewable energy
linear_trend_model_renewable <- lm(energy_data_tsa[,1]~t)
summary(linear_trend_model_renewable)
```

```
##
## Call:
## lm(formula = energy_data_tsa[, 1] ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -154.81  -39.55   12.52   41.49  171.15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.44868    5.11085   33.55  <2e-16 ***
## t           0.74999     0.01397   53.69  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.22 on 631 degrees of freedom
## Multiple R-squared:  0.8204, Adjusted R-squared:  0.8201
## F-statistic: 2883 on 1 and 631 DF, p-value: < 2.2e-16
```

```
#Fit a linear trend to TS of hydroelectric energy
linear_trend_model_hydroelectric <- lm(energy_data_tsa[,2]~t)
summary(linear_trend_model_hydroelectric)
```

```
##
## Call:
## lm(formula = energy_data_tsa[, 2] ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.190 -10.214  -0.715    8.909   39.723
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 83.223802    1.110552   74.939  < 2e-16 ***
## t          -0.012199    0.003035   -4.019 6.55e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.95 on 631 degrees of freedom
## Multiple R-squared:  0.02496, Adjusted R-squared:  0.02342
## F-statistic: 16.15 on 1 and 631 DF, p-value: 6.547e-05
```

```
#Store reg coefficient for renewable energy production
beta0_renewable <- as.numeric(linear_trend_model_renewable$coefficients[1]) #intercept
beta1_renewable <- as.numeric(linear_trend_model_renewable$coefficients[2]) #slope

#Store reg coefficient for hydroelectric energy production
beta0_hydro <- as.numeric(linear_trend_model_hydroelectric$coefficients[1]) #intercept
beta1_hydro <- as.numeric(linear_trend_model_hydroelectric$coefficients[2]) #intercept
```

Answer: Renewable Energy Production: The p value is smaller than 0.05, which is a statistical significance and shows a positive upward trend. Starting from 171.4 TBtu, one unit increase in time period (month) will result 0.75 TBtu increases in total renewable energy production.

Hydroelectric Energy Consumption: The p value is smaller than 0.05, which is a statistical significance and shows a very small and negative downward trend. Starting from 83.2 TBtu, one unit increase in time period (month) will result 0.01 TBtu decreases in total renewable energy production.

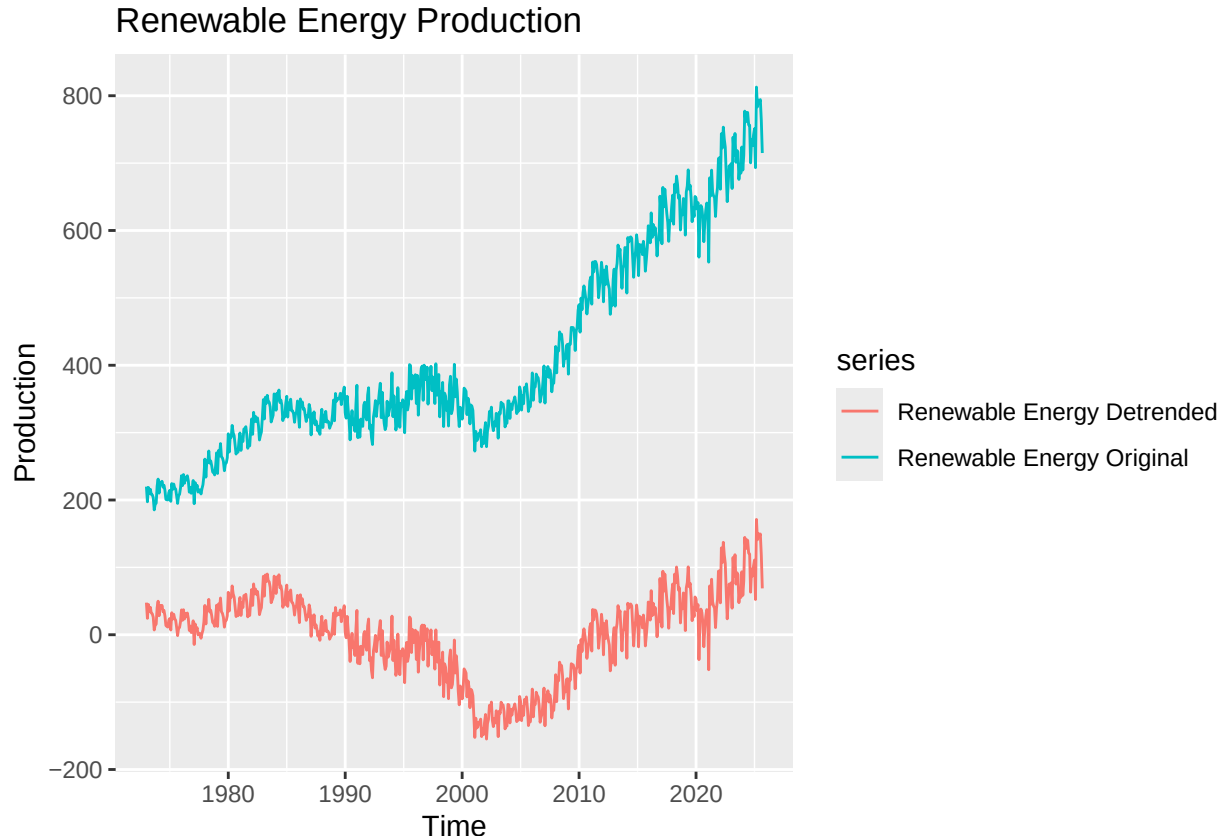
Q4

Use the regression coefficients to detrend each series (subtract fitted linear trend). Plot detrended series and compare with the original time series from Q1. Describe what changed.

```
#remove the trend from series for renewable energy production
detrend_renewable <- energy_data_tsa[,1] - (beta0_renewable + beta1_renewable*t)
class(detrend_renewable) #test if it is a time series format
```

```
## [1] "ts"
```

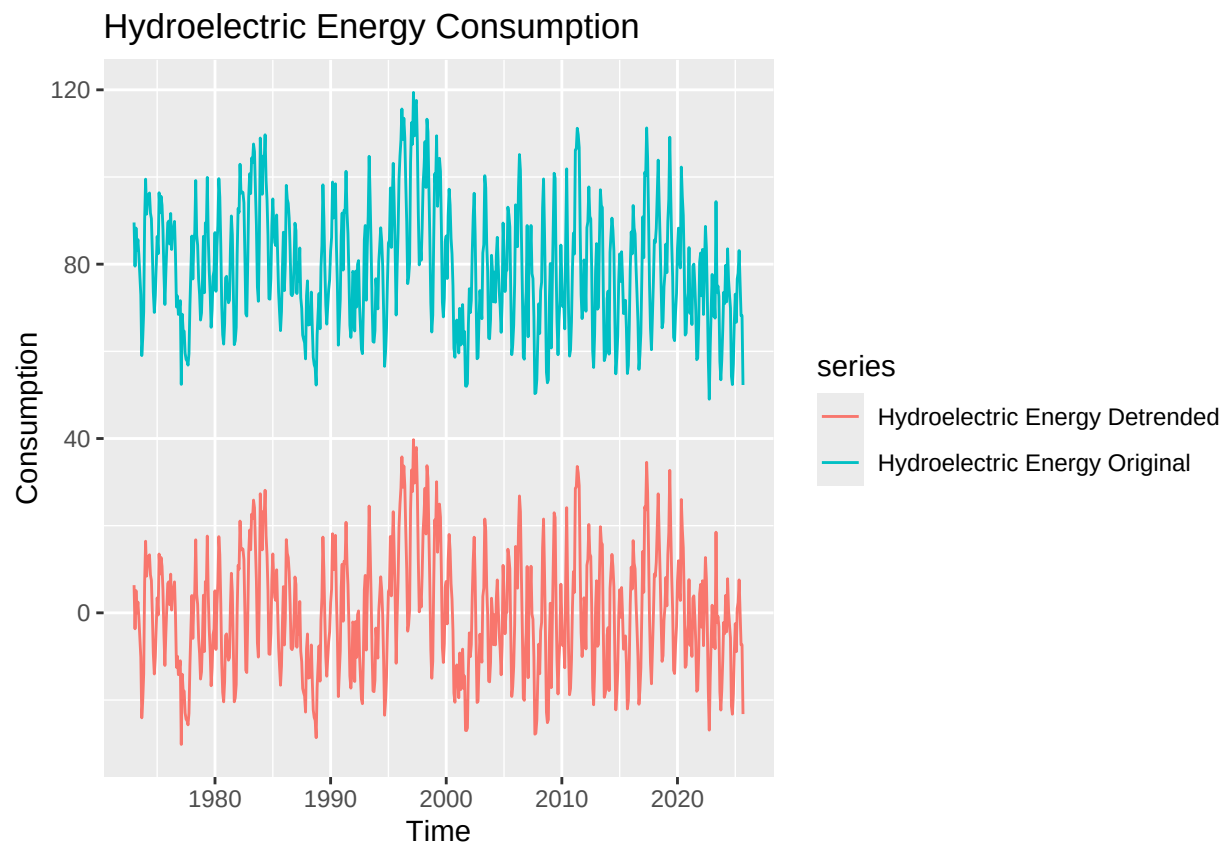
```
#Plot detrended series for renewable energy production
autoplot(energy_data_tsa[, 1], series = "Renewable Energy Original") +
  autolayer(detrend_renewable, series = "Renewable Energy Detrended") +
  labs(
    title = "Renewable Energy Production",
    x = "Time",
    y = "Production",
    colour = "series" )
```



```
#remove the trend from series for hydroelectric energy consumption
detrend_hydro <- energy_data_tsa[,2] - (beta0_hydro + beta1_hydro*t)
class(detrend_hydro) #test if it is a time series format
```

```
## [1] "ts"
```

```
#Plot detrended series for renewable energy production
autoplot(energy_data_tsa[, 2], series = "Hydroelectric Energy Original") +
  autolayer(detrend_hydro, series = "Hydroelectric Energy Detrended") +
  labs(
    title = "Hydroelectric Energy Consumption",
    x = "Time",
    y = "Consumption",
    colour = "series" )
```



> Answer: After being detrended, the renewable energy production does not show a clear upward trend anymore. Instead, it fluctuates around 0 (a constant level). The hydroelectric consumption series after detrend looks very similar to the original one: fluctuating around a constant level. The difference is that the constant level moves from 80 TBtu to 0 TBtu.

Q5

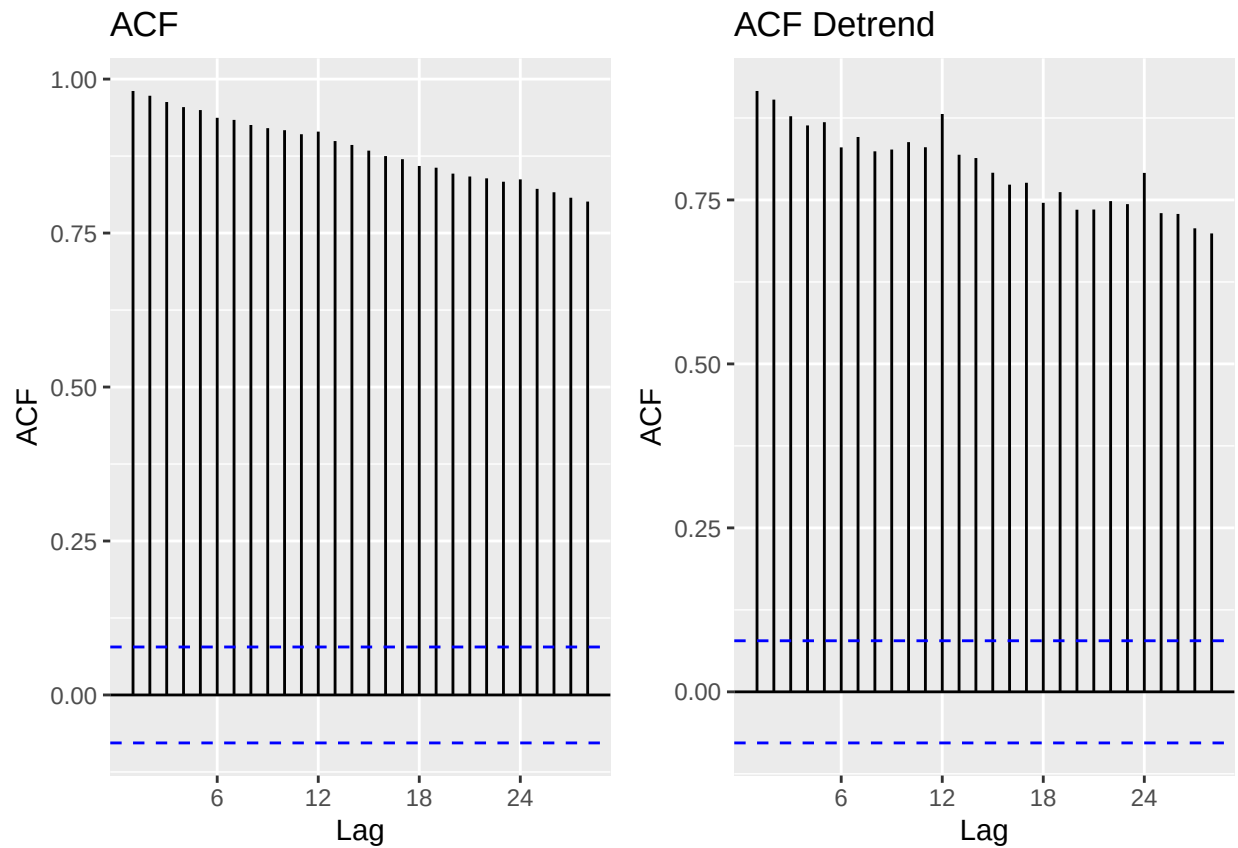
Plot ACF and PACF for the detrended series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side to make it easier to compare. Did the plots change? How?

```
#Plot original and detrended Renewable ACF
```

```
Renewable_acf_detrend <- ggAcf(detrend_renewable) + ggtitle("ACF Detrend")
```

```
Renewable_Original_Detrend_combined_ACF <- plot_grid(Renewable_acf, Renewable_acf_detrend, nrow = 1)
```

```
Renewable_Original_Detrend_combined_ACF
```

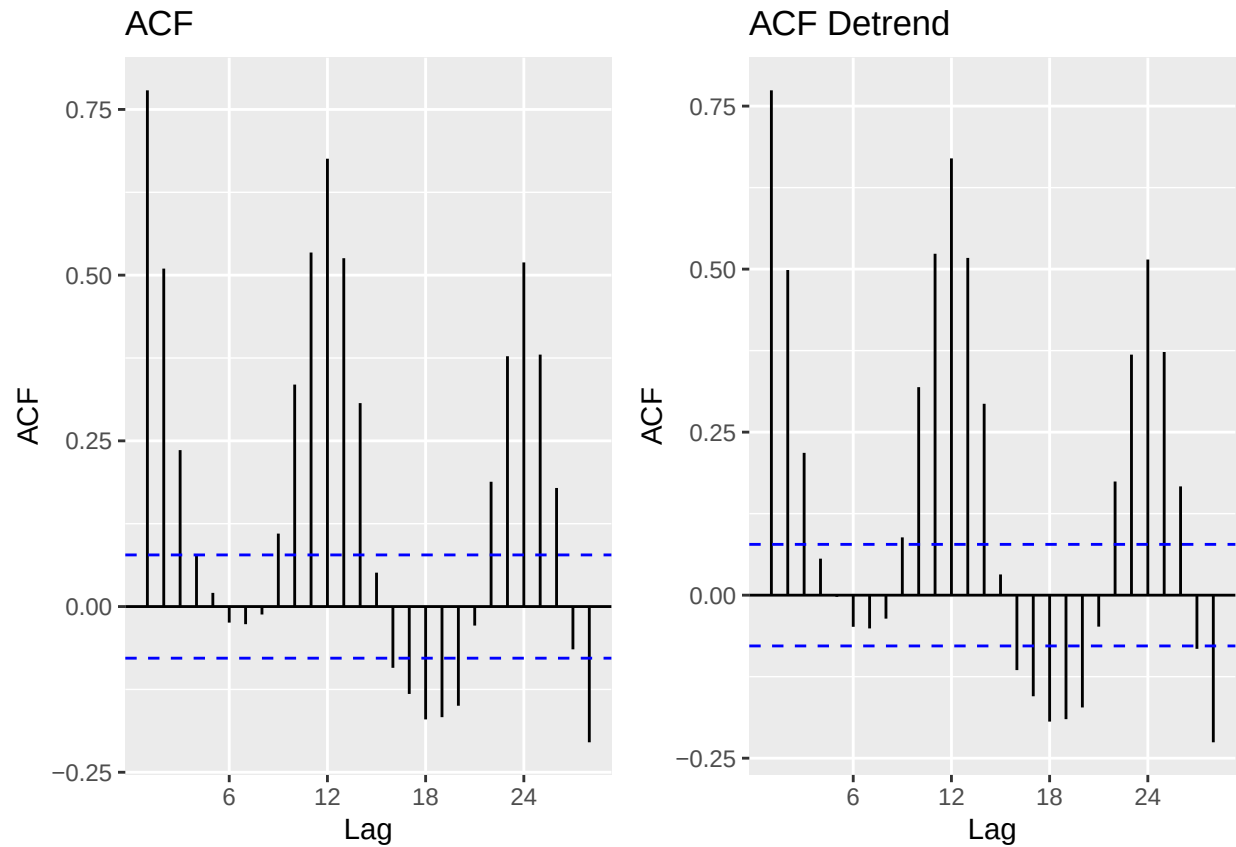


```
#Plot original and detrended Hydroelectric ACF
```

```
Hydro_acf_detrend <- ggAcf(detrend_hydro) + ggtitle("ACF Detrend")
```

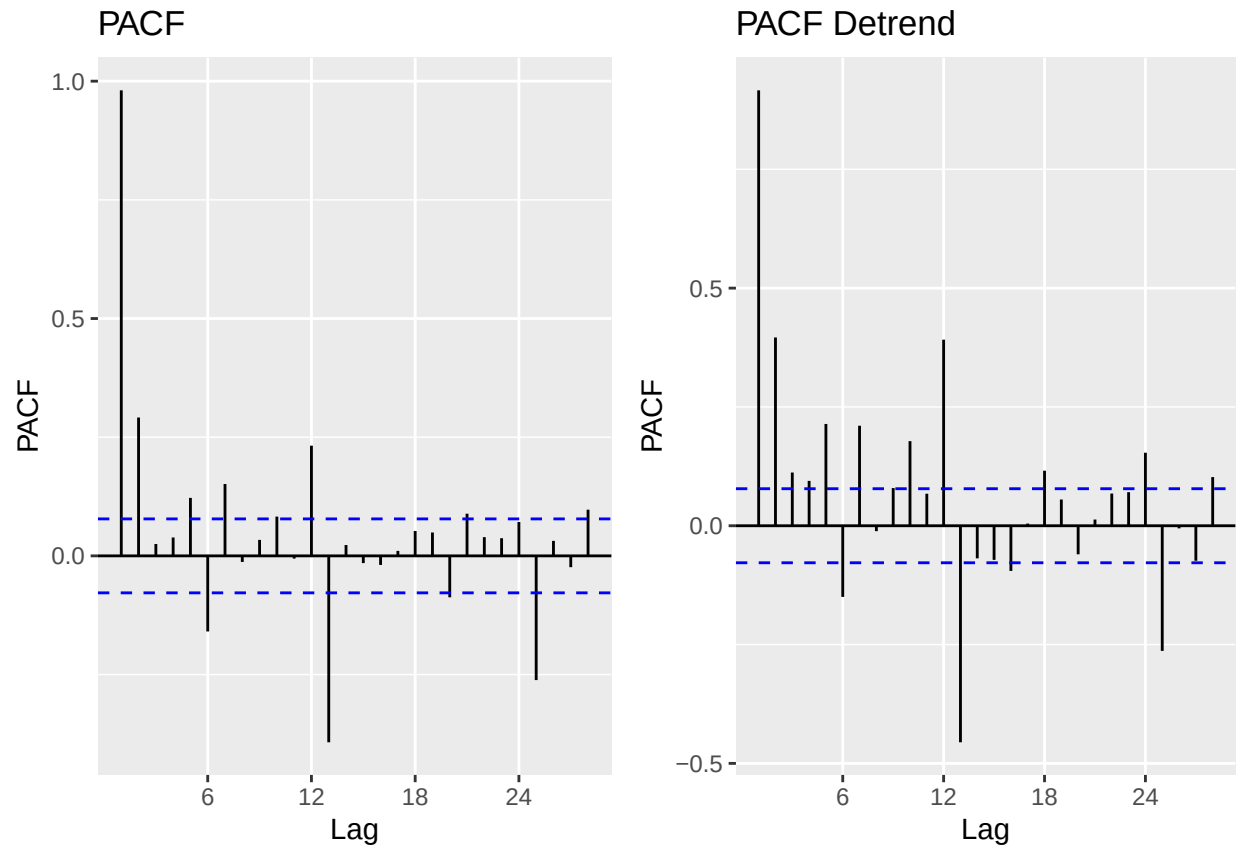
```
Hydro_Original_Detrend_combined_ACF <- plot_grid(Hydro_acf, Hydro_acf_detrend, nrow = 1)
```

```
Hydro_Original_Detrend_combined_ACF
```



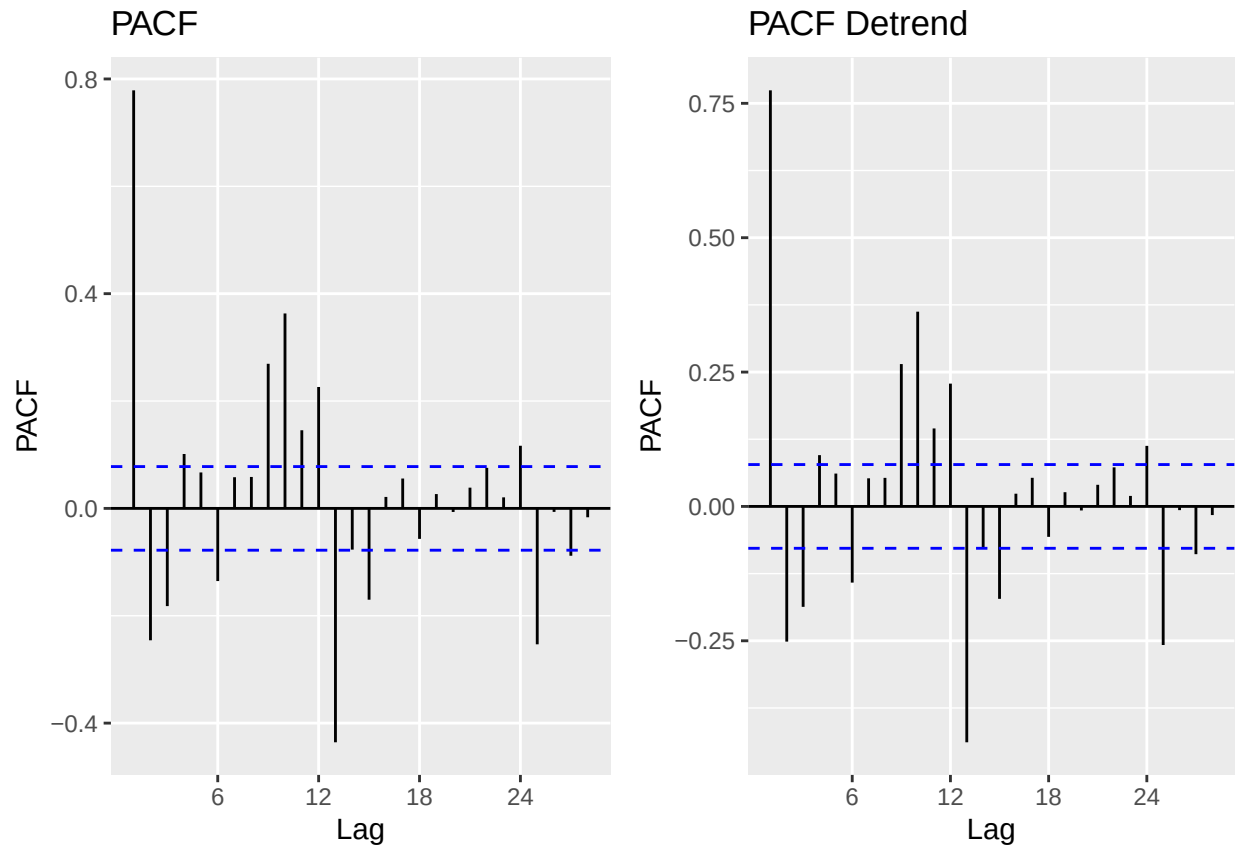
```
#Plot original and detrended Renewable PACF
Renewable_Pacf_detrend <- ggPacf(detrend_renewable) + ggtitle("PACF Detrend")

Renewable_Original_Detrend_combined_PACF <- plot_grid(Renewable_Pacf, Renewable_Pacf_detrend, nrow = 1)
Renewable_Original_Detrend_combined_PACF
```



```
#Plot original and detrended Hydroelectric ACF
Hydro_Pacf_detrend <- ggPacf(detrend_hydro) + ggtitle("PACF Detrend")

Hydro_Original_Detrend_combined_PACF <- plot_grid(Hydro_Pacf, Hydro_Pacf_detrend, nrow = 1)
Hydro_Original_Detrend_combined_PACF
```



> Answer: After being detrended, the renewable energy production's ACF is still high and positive, while it decays a bit faster and spikes at some lags. The change indicates that linear trend is a main source that explains correlation. The renewable energy production's PACF has a significant value at lag 1, showing a strong persistence. The PACF of renewable energy production looks similar before and after detrend, while some spikes become more apparent after removing linear trend, showing that the renewable energy production has an annual cycle. The hydroelectric energy consumption's ACF and PACF do not change after removing the linear trend, indicating that the linear trend does not impact the correlation. The trends can be seasonal.

Seasonal Component

Set aside the detrended series and consider the original series again from Q1 to answer Q6 to Q8.

Q6

Just by looking at the time series and the acf plots, do the series seem to have a seasonal trend? No need to run any code to answer your question. Just type in your answer below.

Answer: The renewable energy production does not seem to have a season trend, while the hydroelectric consumption shows an apparent seasonal trend. The renewable energy production's ACF decays slowly from a large correlation value, showing a strong persistence instead of seasonality. The hydroelectric consumption time series fluctuates around a constant level, and the acfs before and after detrend spike every 12 lags, indicating a strong seasonality instead of linear trend.

Q7

Use function `lm()` to fit a seasonal means model (i.e. using the seasonal dummies) to the two time series. Ask R to print the summary of the regression. Interpret the regression output. From the results, which series have a seasonal trend? Do the results match you answer to Q6?

```
#Use seasonal means model
#dummies (renewable energy)
dummies_renewable <- seasonaldummy(energy_data_tsa[,1])

seas_means_model_renewable <- lm(energy_data_tsa[,1]~dummies_renewable)
summary(seas_means_model_renewable)
```

```
##
## Call:
## lm(formula = energy_data_tsa[, 1] ~ dummies_renewable)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -213.33  -97.36  -59.88   121.55   389.62
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      417.265      21.096   19.779  <2e-16 ***
## dummies_renewableJan      2.090      29.693    0.070   0.944
## dummies_renewableFeb    -34.524      29.693   -1.163   0.245
## dummies_renewableMar     5.956      29.693    0.201   0.841
## dummies_renewableApr    -6.900      29.693   -0.232   0.816
## dummies_renewableMay     8.162      29.693    0.275   0.784
## dummies_renewableJun    -2.231      29.693   -0.075   0.940
## dummies_renewableJul     3.864      29.693    0.130   0.897
## dummies_renewableAug    -3.978      29.693   -0.134   0.893
## dummies_renewableSep   -29.033      29.693   -0.978   0.329
## dummies_renewableOct   -19.937      29.834   -0.668   0.504
## dummies_renewableNov   -20.617      29.834   -0.691   0.490
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 152.1 on 621 degrees of freedom
## Multiple R-squared:  0.008243, Adjusted R-squared: -0.009324
## F-statistic: 0.4692 on 11 and 621 DF, p-value: 0.9223
```

```
#store coeff
beta_int_renewable <- seas_means_model_renewable$coefficients[1]
beta_coeff_renewable <- seas_means_model_renewable$coefficients[2:12]

#dummies (hydroelectric)
dummies_hydro <- seasonaldummy(energy_data_tsa[,2])
seas_means_model_hydro <- lm(energy_data_tsa[,2]~dummies_hydro)
summary(seas_means_model_hydro)
```

```
##
## Call:
```

```
## lm(formula = energy_data_tsa[, 2] ~ dummies_hydro)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.895  -6.368  -0.595   6.213  32.557
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      79.724      1.436  55.511 < 2e-16 ***
## dummies_hydroJan    4.951      2.021   2.449  0.014591 *
## dummies_hydroFeb   -2.415      2.021  -1.195  0.232608
## dummies_hydroMar    7.116      2.021   3.520  0.000463 ***
## dummies_hydroApr    5.614      2.021   2.777  0.005649 **
## dummies_hydroMay   14.080      2.021   6.965  8.38e-12 ***
## dummies_hydroJun   10.780      2.021   5.333  1.36e-07 ***
## dummies_hydroJul    4.003      2.021   1.980  0.048091 *
## dummies_hydroAug   -5.320      2.021  -2.632  0.008710 **
## dummies_hydroSep  -16.598      2.021  -8.211  1.28e-15 ***
## dummies_hydroOct  -16.329      2.031  -8.040  4.56e-15 ***
## dummies_hydroNov  -10.782      2.031  -5.308  1.54e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.36 on 621 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4621
## F-statistic: 50.35 on 11 and 621 DF, p-value: < 2.2e-16
```

```
#store coeff
beta_int_hydro <- seas_means_model_hydro$coefficients[1]
beta_coeff_hydro <- seas_means_model_hydro$coefficients[2:12]
```

Answer: The p-value for renewable energy production's seasonality is 0.9223, larger than 0.05, fail to reject the null hypothesis that it does not have seasonal trend. All p-values in all months do not show significance, matching with the answer to Q6 that total renewable energy production has linear trend instead of seasonality. The p-value for hydroelectric energy consumption is very small, reject the null hypothesis, indicating that it contains strong seasonal trend, which also matches the answer in Q6.

Q8

Use the regression coefficients from Q7 to deseason the series. Plot the deseason series and compare with the plots from part Q1. Did anything change?

Renewable:

```
#compute seas component
season_renewable_data <- array(0,nobs)
for(i in 1:nobs) {
  season_renewable_data[i] <- beta_int_renewable+beta_coeff_renewable%*%dummies_renewable[i,]
}

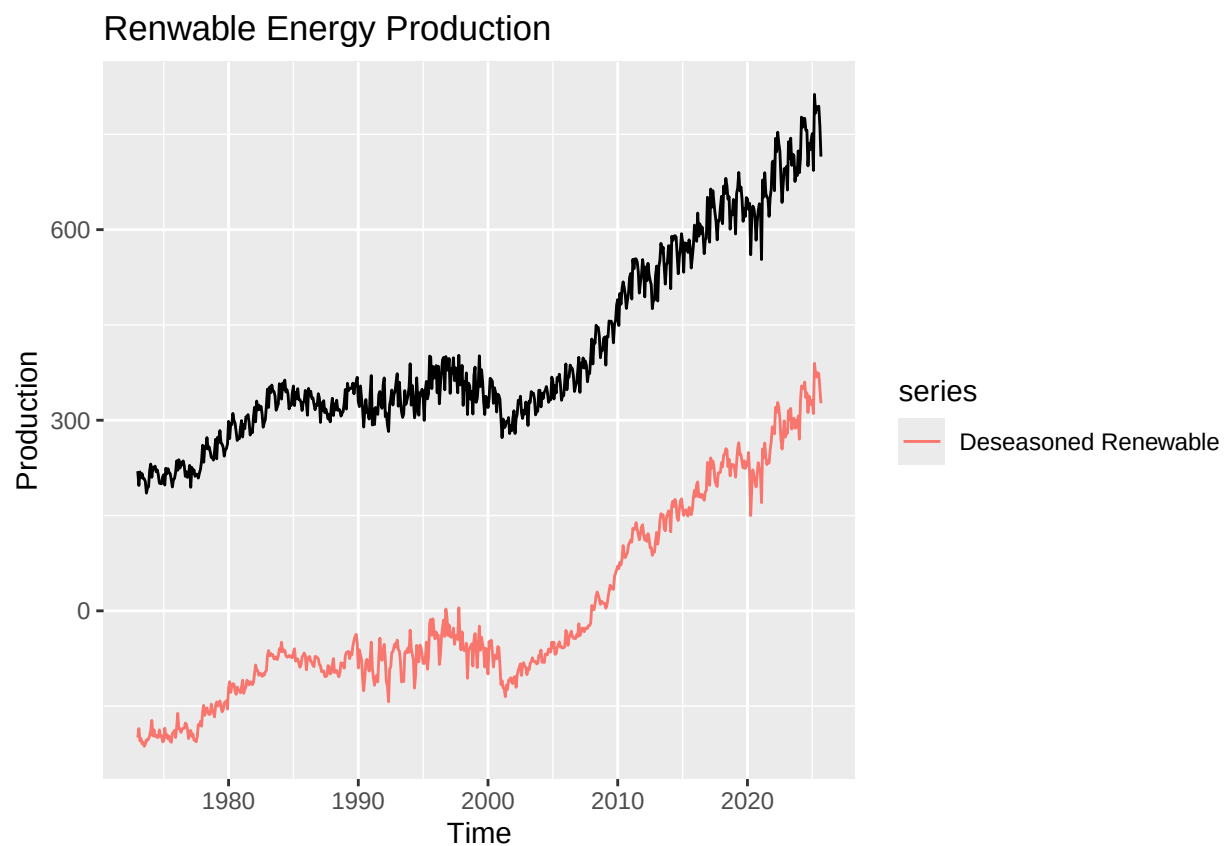
#removing seas component from series
deseason_renewable_data <- energy_data_tsa[,1] - season_renewable_data
```

```

#convert to ts obj
ts_deseason_renewable_data <- ts(deseason_renewable_data,
                                start=c(1973,1),
                                frequency=12)

#plots
autoplot(energy_data_tsa[,1])+
  autolayer(ts_deseason_renewable_data, series= "Deseasoned Renewable") +
  labs(
    title = "Renwable Energy Production",
    x = "Time",
    y = "Production",
    colour = "series" )

```



Hydroelectric:

```

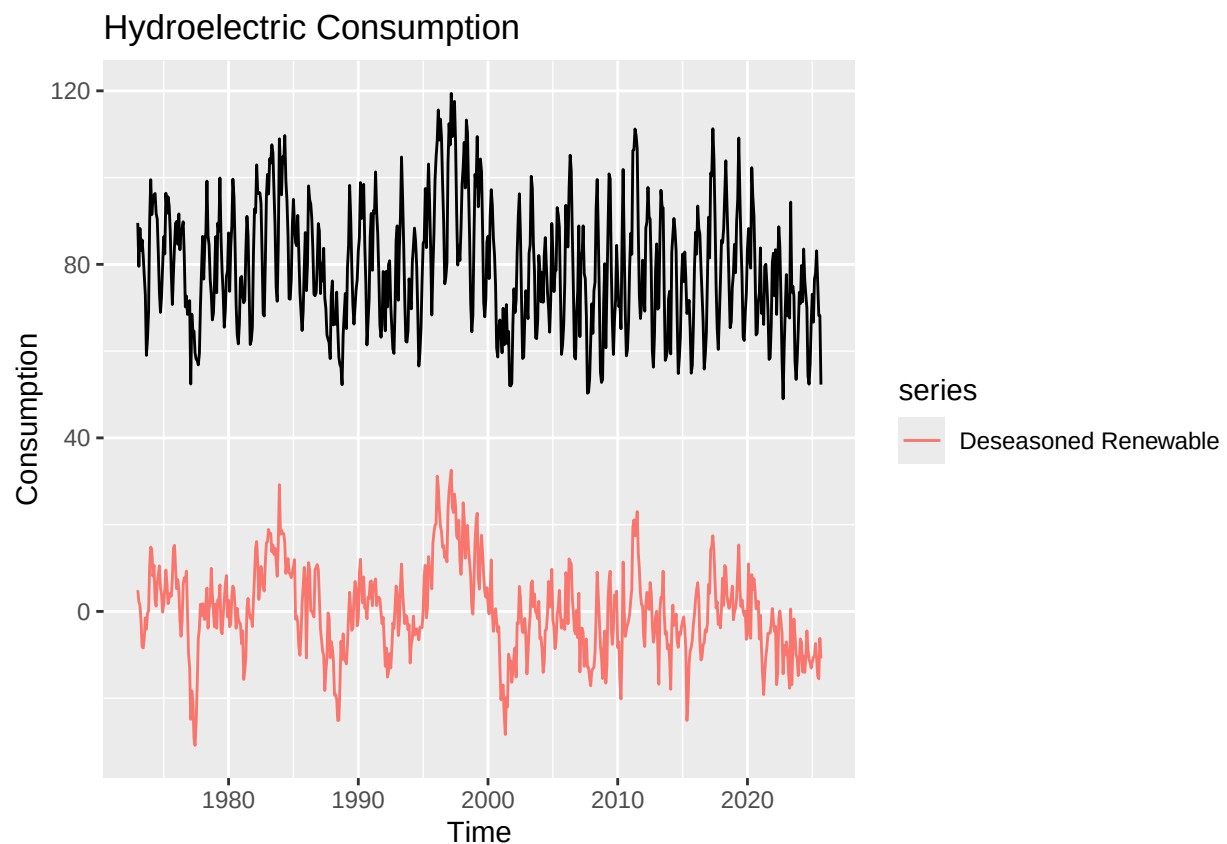
#compute seas component
season_hydro_data <- array(0,nobs)
for(i in 1:nobs) {
  season_hydro_data[i] <- beta_int_hydro+beta_coeff_hydro*%dummies_hydro[i,]
}

#removing seas compoment from series
deseason_hydro_data <- energy_data_tsa[,2] - season_hydro_data

```

```
#convert to ts obj
ts_deseason_hydro_data <- ts(deseason_hydro_data,
                             start=c(1973,1),
                             frequency=12)

#plots
autoplot(energy_data_tsa[,2])+
  autolayer(ts_deseason_hydro_data, series="Deseasoned Renewable") +
  labs(
    title = "Hydroelectric Consumption",
    x = "Time",
    y = "Consumption",
    colour = "series" )
```



Answer: The total renewable energy production plot looks very similar to the original one, meaning that seasonality is not a main source of the correlation. The hydroelectric consumption plot has some changes and the fluctuations become smoother, indicating that seasonality is a main source of the correlation.

Q9

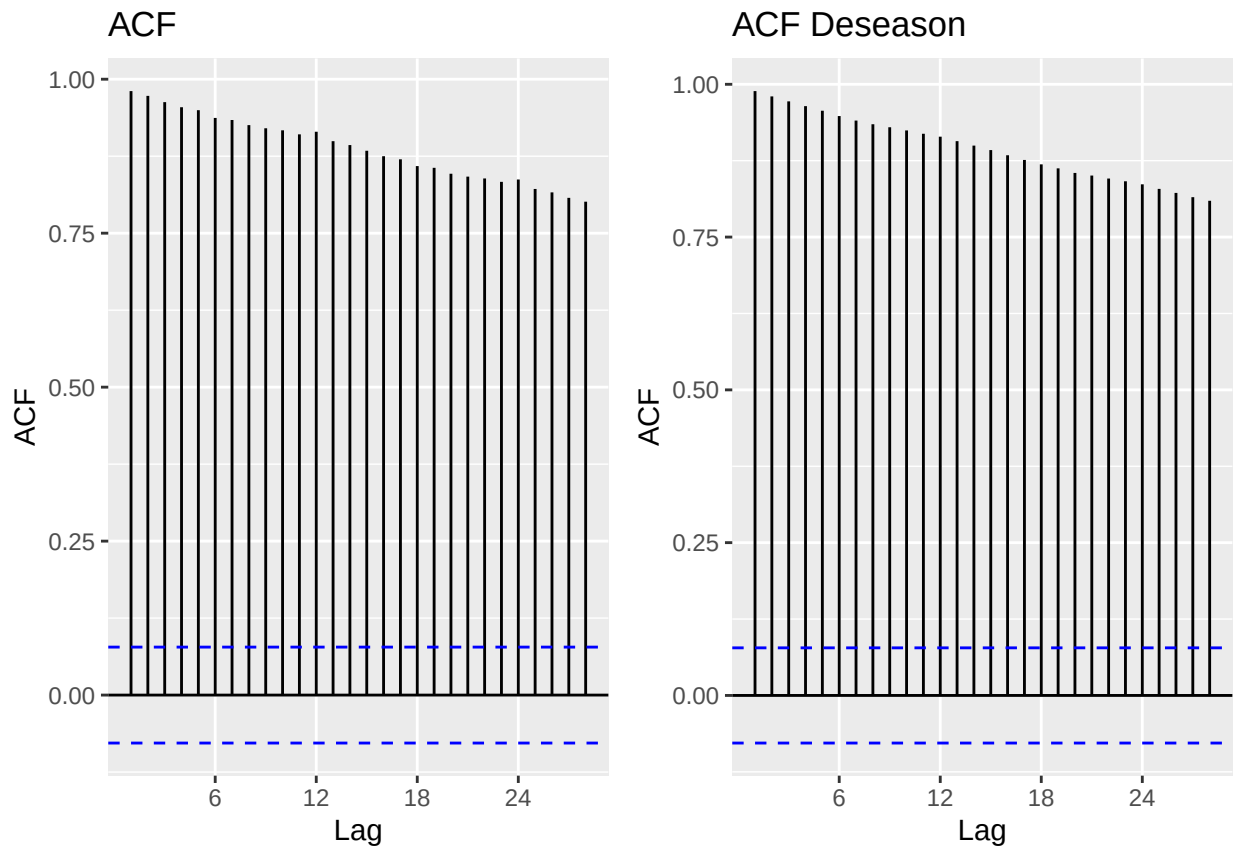
Plot ACF and PACF for the deseason series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side. Did the plots change? How?

```
#Plot original and deseasoned Renewable ACF
```

```
Renewable_acf_deseason <- ggAcf(ts_deseason_renewable_data) + ggtitle("ACF Deseason")
```

```
Renewable_Original_Deseason_combined_ACF <- plot_grid(Renewable_acf, Renewable_acf_deseason, nrow = 1)
```

```
Renewable_Original_Deseason_combined_ACF
```

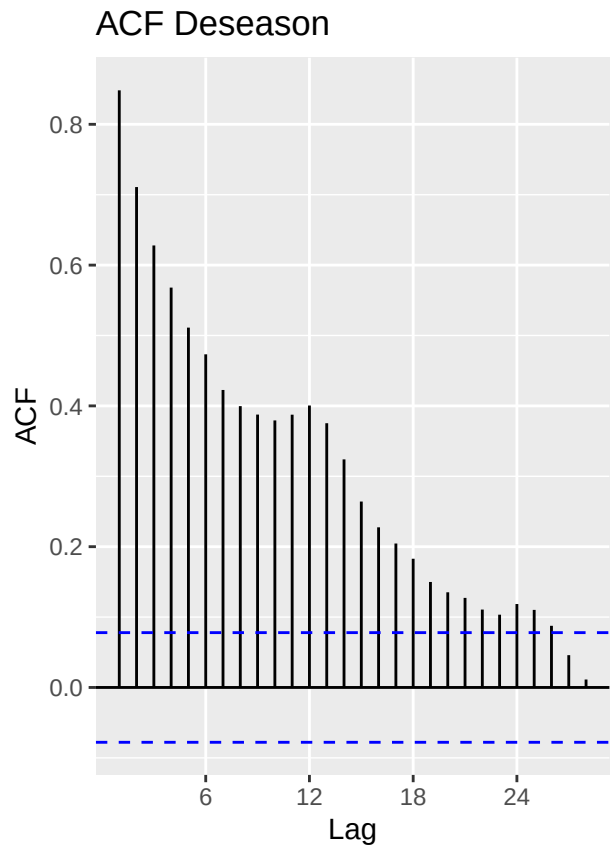
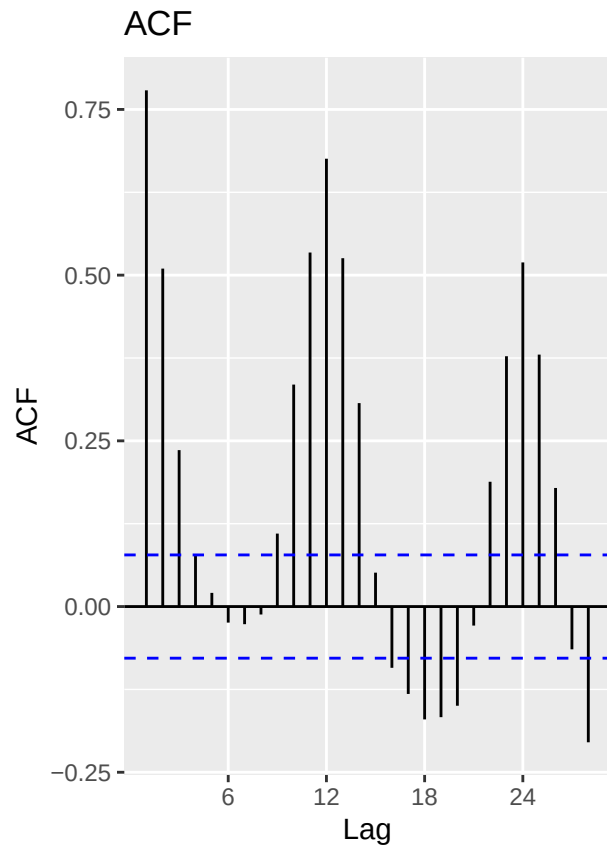


```
#Plot original and deseasoned Hydroelectric ACF
```

```
Hydro_acf_deseason <- ggAcf(ts_deseason_hydro_data) + ggtitle("ACF Deseason")
```

```
Hydro_Original_Deseason_combined_ACF <- plot_grid(Hydro_acf, Hydro_acf_deseason, nrow = 1)
```

```
Hydro_Original_Deseason_combined_ACF
```

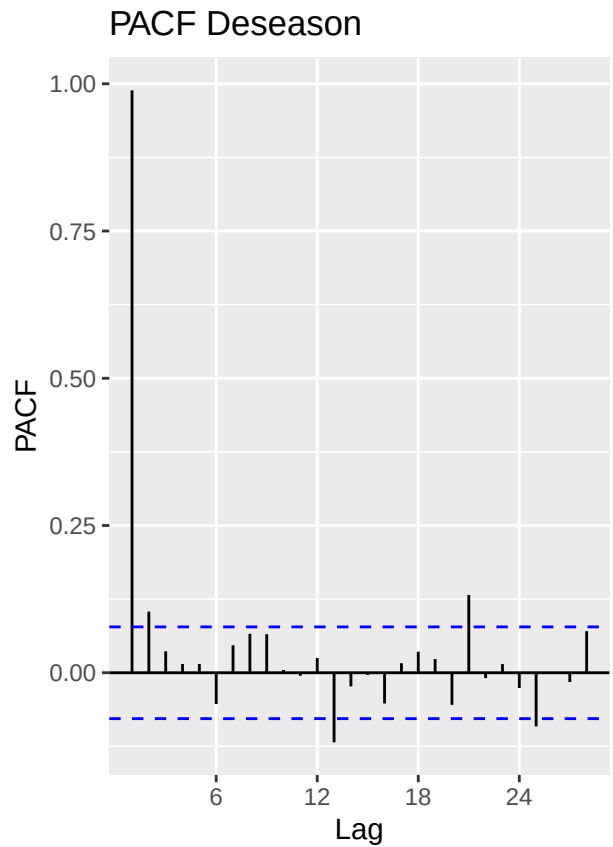
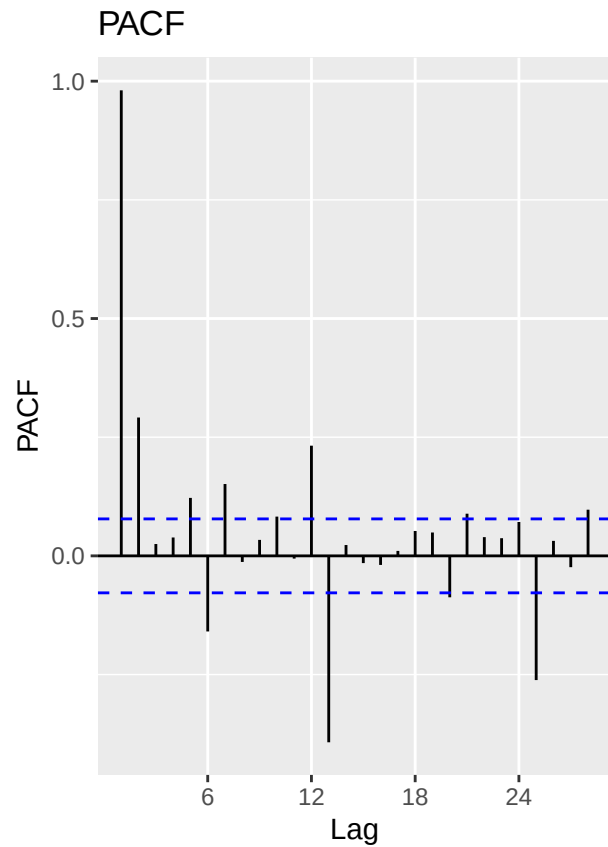


```
#Plot original and deseasoned Renewable ACF
```

```
Renewable_Pacf_deseason <- ggPacf(ts_deseason_renewable_data) + ggtitle("PACF Deseason")
```

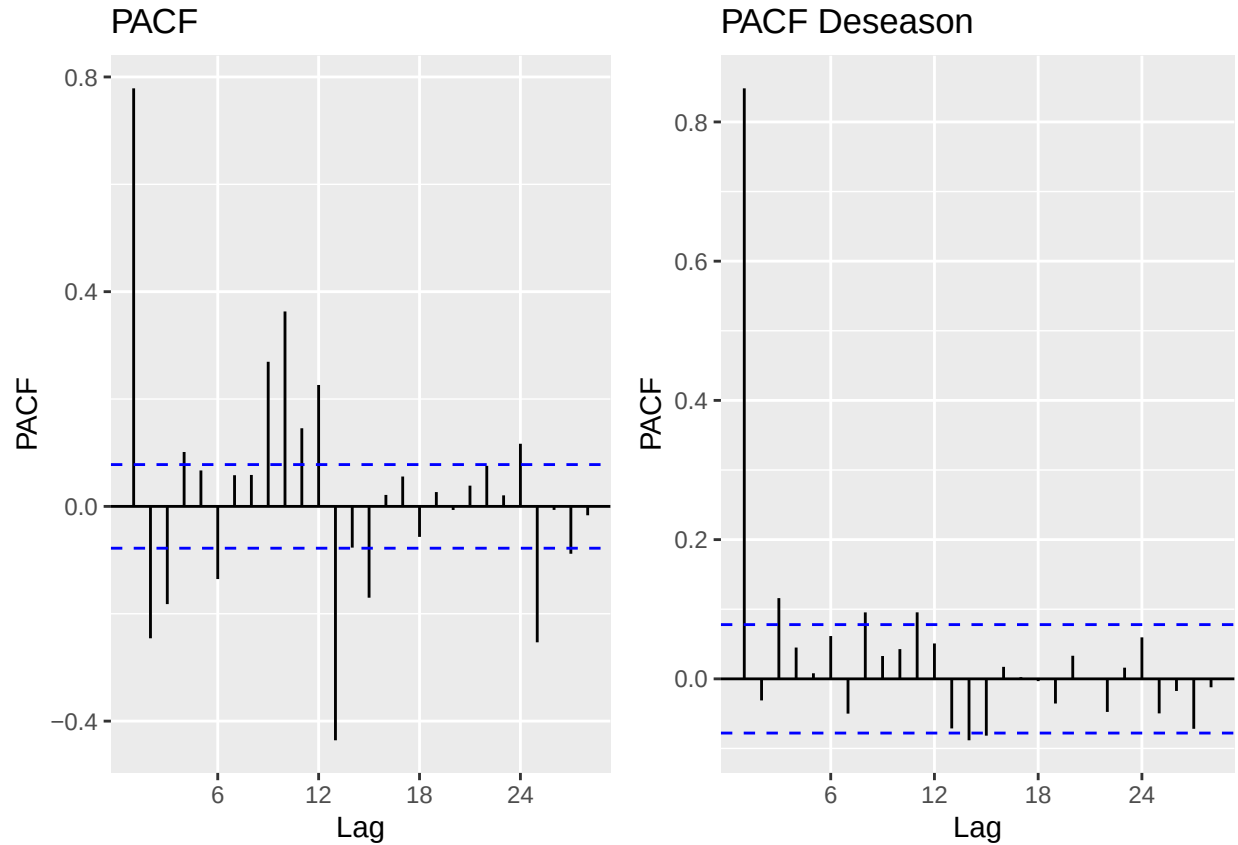
```
Renewable_Original_Deseason_combined_PACF <- plot_grid(Renewable_Pacf, Renewable_Pacf_deseason, nrow = 2)
```

```
Renewable_Original_Deseason_combined_PACF
```



```
#Plot original and deseasoned Hydroelectric ACF
Hydro_Pacf_deseason  <- ggPacf(ts_deseason_hydro_data) + ggtitle("PACF Deseason")

Hydro_Original_Deseason_combined_PACF <- plot_grid(Hydro_Pacf, Hydro_Pacf_deseason, nrow = 1)
Hydro_Original_Deseason_combined_PACF
```



> Answer: The total renewable energy production's ACFs look very similar before and after removing seasonality, which means that seasonality does not have a significant effect on the long-run persistence. Same as the PACF. Deseasoned PACF of renewable energy production still shows similar spikes as the original one. The hydroelectric consumption's ACF changes before and after deseasoning. The deseasoned ACF does not have fluctuations and spikes every 12 lags. Instead, the ACF slowly decays from a large value (close to 1) to negative, indicating the strong seasonality of hydroelectric consumption. The PACF also changes after deseasoning. In conclusion, the main source of correlation for renewable energy production is linear trend, while that of the hydroelectric consumption is seasonality.